Qcombo: 自动计算多体算符对易式的Python软件包

L. H. Chen[a,b], Y. Li[a,b], H. Hergert[c,d], J. M. Yao[a,b,*]

[a] *School of Physics and Astronomy, Sun Yat-sen University, Zhuhai 519082, P.R. China*
[b] *Guangdong Provincial Key Laboratory of Quantum Metrology and Sensing, Sun Yat-Sen University, Zhuhai 519082, P.R. China*
[c] *Facility for Rare Isotope Beams, Michigan State University, East Lansing, Michigan 48824-1321, USA*
[d] *Department of Physics & Astronomy, Michigan State University, East Lansing, Michigan 48824-1321, USA*

Chen Lin Hai

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Abstract

Abstract

qcombo is a Python package for the symbolic evaluation of commutators between general quantum many-body operators expressed in normal-ordered form using the generalized Wick theorem. The package provides an automated and systematic framework for generating the corresponding algebraic expressions, significantly reducing the risk of human error in lengthy and complex analytical derivations. It is designed to assist the development and implementation of modern many-body methods in nuclear physics, quantum chemistry, and related fields. The functionality and workflow of the package are demonstrated through an application to the in-medium similarity renormalization group (IMSRG) method, which has been widely used for nuclear ab initio calculations. As a representative example, qcombo is employed to automatically generate the complete set of multi-reference IMSRG flow equations with operators truncated at the normal-ordered three-body level.

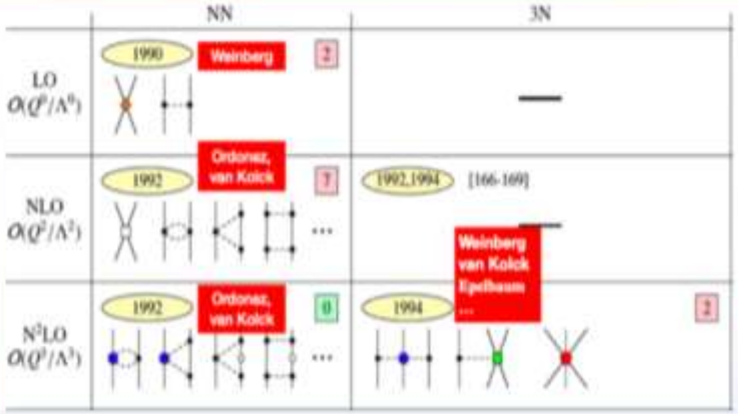
Qcombo is a Python package for automatedly computing and simplifying the commutator expression of normal-ordered many-body operators based on generalized Wick theorem.

Introduction

- **原子核从头计算方法（Ab Initio Methods）：** 基于手征有效场论构建的核力，（近似）严格求解原子核多体问题的理论框架。

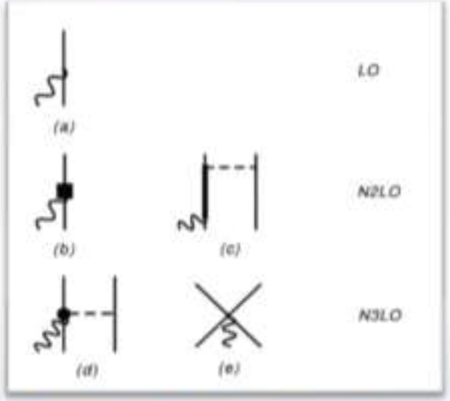
  - Nuclear forces & transition operators: consistently constructed within **chiral EFT** (N, π)
  - The theoretical uncertainty: controlled by the power-counting $(Q/\Lambda)^n$ with Q ≈ 140 MeV, Λ ≈ 700 MeV)
  - Nuclear wave functions: solved using **exact or systematically improvable many-body methods**

Nuclear forces



Phys. Rep. 503, 1 (2011); 890,1 (2020)

Axial currents



Phys. Rev. C 95, 059902 (2017)
Few Body Syst. 62, 114 (2021)

- **Nuclear ab initio methods**
  - ✓ 无芯壳模型 (NCSM)
  - ✓ 介质相似重整化群方法 (IMSRG)
  - ✓ 格林函数蒙特卡罗方法 (GFMC)
  - ✓ 耦合团簇理论 (CC)
  - ✓ 多体微扰论方法 (MBPT)
  - ✓ Brueckner Hartree-Fock (BHF)
  - ✓ …

Frontiers in Physics 8, 379 (2020)

IMSRG

- **基本思想: 通过引入连续的幺正变换U(s)将H逐渐对角化**

$$H(s) = U(s)H(0)U^{\dagger}(s)$$

$$\frac{dH(s)}{ds} = \frac{dU(s)}{ds}U^{\dagger}(s)\,H(s) + H(s)U(s)\frac{dU^{\dagger}(s)}{ds}$$

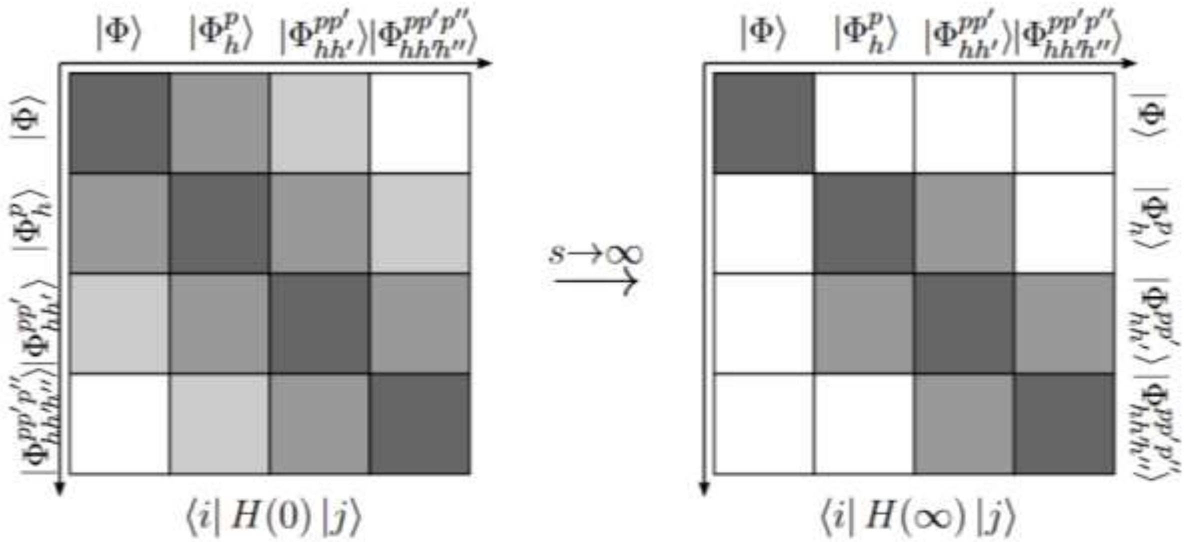
生成元：
(generator)

$$\eta(s) = \frac{dU(s)}{ds}U^{\dagger}(s) = -\eta^{\dagger}(s)$$

- **流方程 （Flow equation）：**

$$\boldsymbol{\frac{d}{ds}H(s) = [\eta(s), H(s)]}$$

多体算符对易式计算



H. Hergert. Phys. Scr. 92, 023002 (2017)

- 将多体算符表示为正规乘积形式

H. Hergert. Phys. Scr. 92, 023002 (2017)

Normal Ordered Operator $a_i^\dagger a_j \equiv \{a_i^\dagger a_j\} + \overline{a_i^\dagger a_j}$

$$\left\langle \Phi_{\mathrm{ref}} \middle| \left\{ A^{i_1 \ldots i_k}_{j_1 \ldots j_k} \right\} \middle| \Phi_{\mathrm{ref}} \right\rangle = 0$$

Normal-Ordered Hamiltonian

$$H = E_0 + \sum_{kl} f_l^k \{A_l^k\} + \frac{1}{4} \sum_{klmn} \Gamma_{mn}^{kl} \{A_{mn}^{kl}\} + \frac{1}{36} \sum_{ijklmn} W_{lmn}^{ijk} \{A_{lmn}^{ijk}\}$$

$$A^{i_1 \ldots i_k}_{j_1 \ldots j_k} = a^\dagger_{i_1} \ldots a^\dagger_{i_k} a_{j_k} \ldots a_{j_1}$$

正规乘积的k-体算符

$$O^{(k)} = \frac{1}{(k!)^2} \sum_{i_1 \ldots j_k} o^{i_1 \ldots i_k}_{j_1 \ldots j_k} \{A^{i_1 \ldots i_k}_{j_1 \ldots j_k}\}$$

IMSRG(2)

$$\eta(s) \approx \eta^{(1)}(s) + \eta^{(2)}(s) + \eta^{(3)}(s)$$

$$H(s) \approx E(s) + f(s) + \Gamma(s) + W(s)$$

$$\frac{d}{ds}H(s) \approx \frac{d}{ds}E(s) + \frac{d}{ds}f(s) + \frac{d}{ds}\Gamma(s) + \frac{d}{ds}W(s)$$

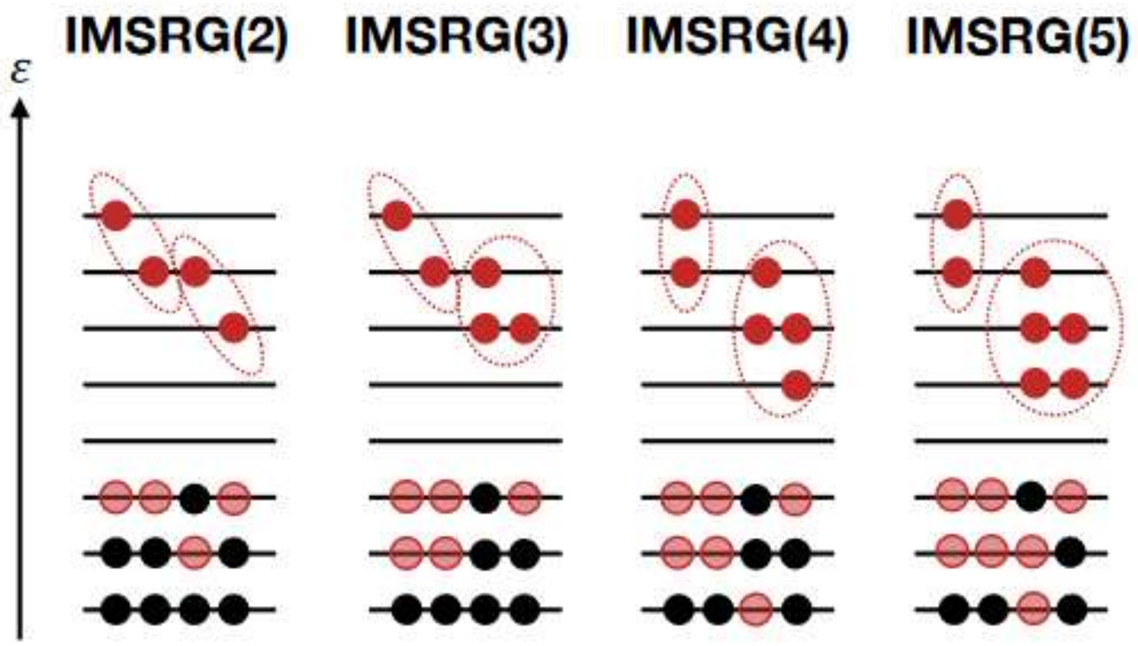
IMSRG(3)

大部分研究采用NO2B近似

推广到开壳原子核：SR-IMSRG(N)

Single Reference(SR)-IMSRG

|Φ⟩: single Slater Determinant



H. Hergert *et al.* J. Phys. Conf. Ser. 1041, 012007 (2018)

- Explicit treatment of **2p–2h (Doubles)** correlations
- **3p–3h (Triples)** and higher → require higher IMSRG truncation
- Recent developments:
  - ➢ **SR-IMSRG(2)**
    - ✓ IMSRG(2*)
    - ✓ IMSRG(3n7)
    - ✓ IMSRG(3f2)
  - ➢ **SR-IMSRG(3)**
  - ➢ **SR-IMSRG(4)**

M. Heinz *et al.* PRC 044318 (2021)

B. C. He *et al.* PRC 044317 (2024)

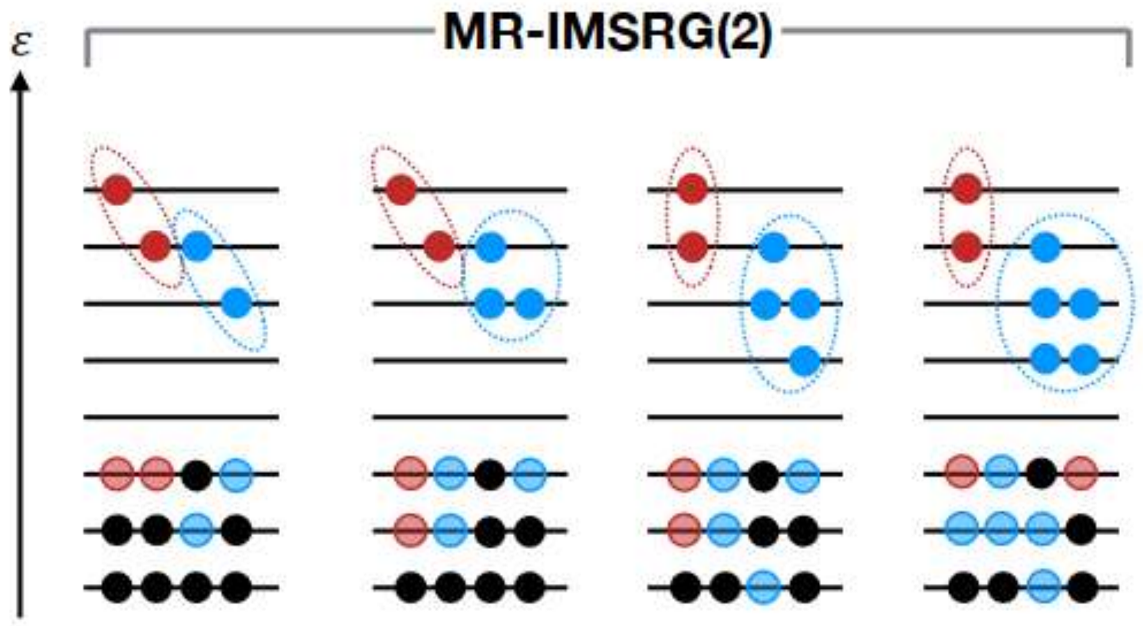
S. R. Stroberg *et al.* PRC 044316 (2024)

R. Z. Hu *et al.* arXiv:2607.08830 (2026)

推广到开壳原子核：MR-IMSRG(2)

Multi-Reference(MR)-IMSRG

$|\Phi\rangle$: Correlated Reference States e.g. PNP $|\,\mathrm{HFB}\rangle$



H. Hergert *et al.* J. Phys. Conf. Ser. 1041, 012007 (2018)

- **Correlated reference state**
- Selected **2p–2h** + all **3p–3h and higher** correlations (built in)
- Recent developments MR-IMSRG(2):

PNP: Hergert (2018)
IM-GCM: Yao (2020)

Extension to MR-IMSRG(3)

Commutators in the MR-IMSRG(3)

- **IMSRG Flow Equation**

$$\frac{d}{ds}H(s) = [\eta(s), H(s)]$$

$$H(s) \approx E(s) + f(s) + \Gamma(s) + W(s)$$

0B 1B 2B 3B

多体算符对易式的计算更加复杂!

$$[G^{(2)}, H^{(3)}]$$

$\cdots$

两体-两体算符之间的对易式(耦合到零体):

$$\begin{aligned}
[G^{(2)}, H^{(2)}]^{(0)} &= \left[\frac{1}{4}\sum_{abcd} G^{ab}_{cd}\{A^{ab}_{cd}\}, \frac{1}{4}\sum_{ijkl} H^{ij}_{kl}\{A^{ij}_{kl}\}\right]^{(0)} \\
&= \sum_{abcd} \frac{1}{4}(n_a n_b \bar{n}_c \bar{n}_d - \bar{n}_a \bar{n}_b n_c n_d) G^{ab}_{cd} H^{cd}_{ab} \\
&\quad + \sum_{abcdef} \left[(n_e - n_f) G^{ae}_{cf} H^{bf}_{de} + \frac{1}{8}(n_e n_f - \bar{n}_e \bar{n}_f)\left(G^{ef}_{cd} H^{ab}_{ef} - G^{ab}_{ef} H^{ef}_{cd}\right)\right] \boxed{\lambda^{ab}_{cd}} \\
&\quad + \sum_{abcdefg} \frac{1}{4}(G^{ag}_{dc} H^{bc}_{fg} - G^{ab}_{dg} H^{cg}_{ef}) \boxed{\lambda^{abc}_{def}}
\end{aligned}$$

多体不可约化密度矩阵

- $\lambda^{(1)}$: $\lambda^i_k \equiv \rho^i_k = \langle \Phi | A^i_k | \Phi \rangle$, $\xi^i_k = \lambda^i_j - \delta^i_j$
- $\lambda^{(2)}$: $\lambda^{ij}_{kl} \equiv \rho^{ij}_{kl} - \mathcal{A}\{\lambda^i_k \lambda^j_l\}$
- $\lambda^{(3)}$: $\lambda^{ijk}_{lmn} \equiv \rho^{ijk}_{lmn} - \mathcal{A}\{\lambda^i_l \lambda^{jk}_{mn}\} - \mathcal{A}\{\lambda^i_l \lambda^j_m \lambda^k_n\}$

Generalized Wick's theorem

SR-IMSRG

$$:A^{a\boxed{b}}_{cd} :: A^{ij}_{\boxed{k}l} : = -\lambda^{\boxed{b}}_{\boxed{k}} :A^{aij}_{cdl}: ,$$

$$:A^{ab}_{\boxed{c}d} :: A^{i\boxed{j}}_{kl} : = -\xi^{\boxed{j}}_{\boxed{c}} :A^{bia}_{dkl}: ,$$

$$:A^{\boxed{ab}}_{cd} :: A^{ij}_{\boxed{kl}} : = +\lambda^{\boxed{ab}}_{\boxed{kl}} :A^{ij}_{cd}: ,$$

$$:A^{a\boxed{b}}_{cd} :: A^{\boxed{i}j}_{\boxed{kl}} : = -\lambda^{\boxed{ib}}_{\boxed{kl}} :A^{aj}_{cd}: ,$$

$$:A^{ab}_{\boxed{c}d} :: A^{\boxed{ij}}_{\boxed{k}l} : = -\lambda^{\boxed{ij}}_{\boxed{ck}} :A^{ab}_{dl}: ,$$

$$:A^{\boxed{ab}}_{c\boxed{d}} :: A^{\boxed{i}j}_{\boxed{kl}} : = -\lambda^{\boxed{abi}}_{\boxed{dkl}} :A^{j}_{c}: ,$$

$$:A^{\boxed{ab}}_{\boxed{cd}} :: A^{\boxed{ij}}_{\boxed{kl}} : = +\lambda^{\boxed{abij}}_{\boxed{cdkl}} .$$

MR-IMSRG

Hergert. H Phys.Scripta 92, 2, 023002 (2017)

irreducible density matrices $\lambda^{(k)}$

- $\lambda^{(1)}$: $\lambda^i_k \equiv \rho^i_k = \langle \Phi | A^i_k | \Phi \rangle$, $\xi^i_k = \lambda^i_j - \delta^i_j$
- $\lambda^{(2)}$: $\lambda^{ij}_{kl} \equiv \rho^{ij}_{kl} - \mathcal{A}\{\lambda^i_k \lambda^j_l\}$
- $\lambda^{(3)}$: $\lambda^{ijk}_{lmn} \equiv \rho^{ijk}_{lmn} - \mathcal{A}\{\lambda^i_l \lambda^{jk}_{mn}\} - \mathcal{A}\{\lambda^i_l \lambda^j_m \lambda^k_n\}$

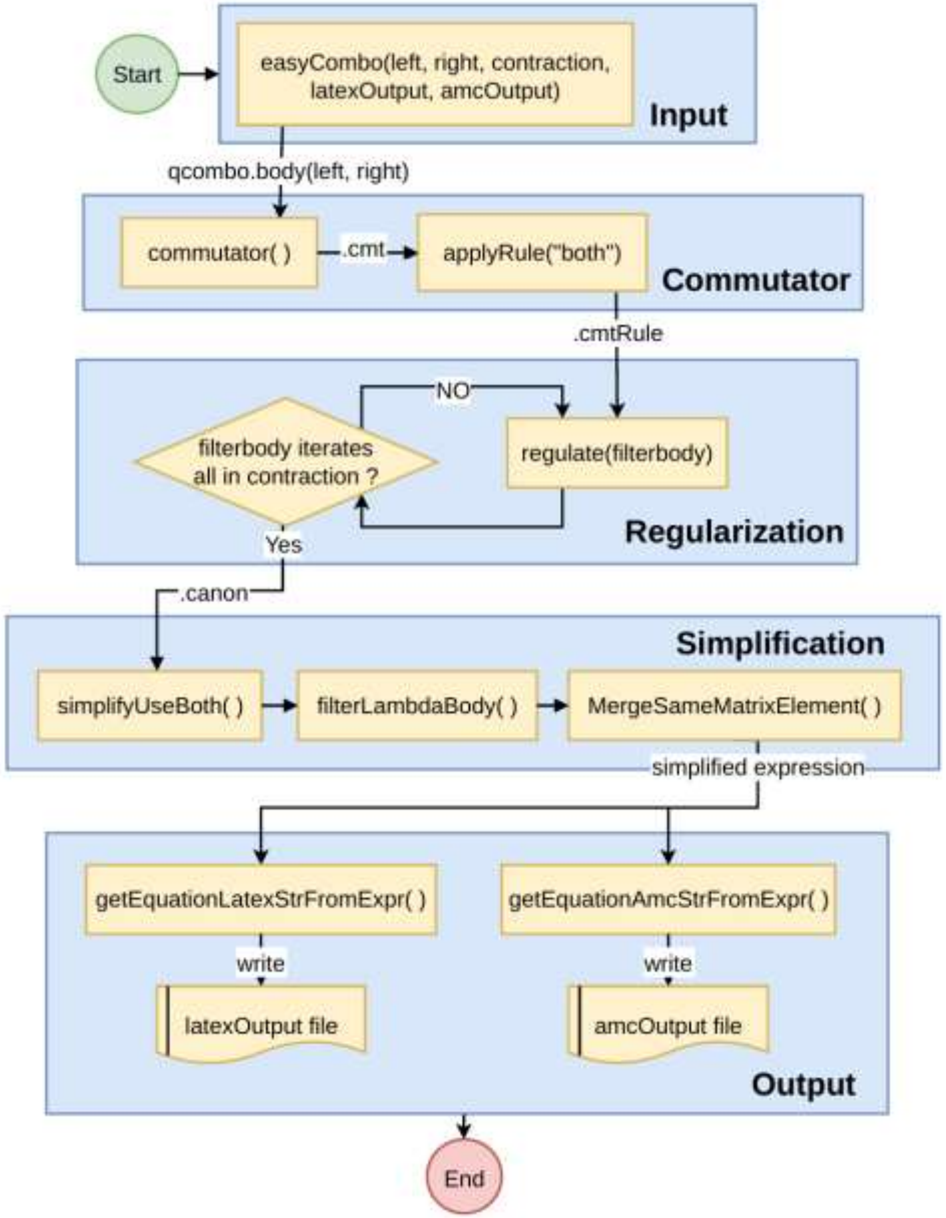
……

where $\mathcal{A}\{\ldots\}$ fully antisymmetrizes the indices of the expression within the brackets, e.g.

$$\mathcal{A}\{\lambda^i_k \lambda^j_l\} = \lambda^i_k \lambda^j_l - \lambda^i_l \lambda^j_k$$

Qcombo

- 从MR-IMSRG(2)拓展到MR-IMSRG(3)需要推导额外的对易式[1B,3B],[2B,3B],[3B,3B]
- 人为推导多体正规序算符的对易式是一个繁琐耗时的过程，尤其是当多体项增加时，人为推导变得费时费力，而且容易犯错。



Qcombo (Quantum-commutator of many-body operators)

- 基于Python的Sympy库搭建的计算多体算符对易式的程序，用于计算任意多体算符的对易式并给出表达式。

Example [1B,2B]

Installation

- The qcombo package is available on PyPI. Users can install qcombo package with pip. (Pyhton ≥ 3.12.0)

```
pip install qcombo
```

- Alternatively, the package can be installed directly from the GitHub repository.

```
git clone https://github.com/chenlh73/qcombo
    .git
cd qcombo
pip install -e .
```

Commutator [1B,2B]

$$[G^{(1)}, H^{(2)}] = \frac{1}{4}\sum_{abijkl} G^a_b H^{ij}_{kl}\big[\{A^a_b\}, \{A^{ij}_{kl}\}\big].$$

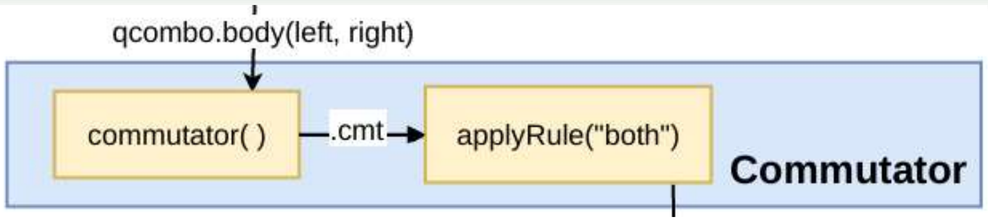
Input

Input module: define left operator and right operator

```
import qcombo
```

```
#[1B,2B]
left=[['a'],['b']]
right=[['i','j'],['k','l']]

#define the bodys
operators = qcombo.bodys(left,right)

# optional settings
operators.wick_mode = "MR"
operators.show_process = True
operators.parallel = False
```

Commutator



Commutator module: Compute $\left[\{A^a_b\},\left\{A^{ij}_{kl}\right\}\right]$

commutator()

```
#compute the commutator
operators.commutator()
#the product
jupyterDisplay(operators.gw)
#the commutator
jupyterDisplay(operators.cmt)
```

.gw :$\{A^a_b\}\{A^{ij}_{kl}\}$

```
output:
```

$$A^{ai}_{kl}\xi^j_b + A^{aij}_{bkl} - A^{aj}_{kl}\xi^i_b - A^a_k\lambda^{ij}_{bl} + A^a_l\lambda^{ij}_{bk} + A^{ij}_{bk}\lambda^a_l - A^{ij}_{bl}\lambda^a_k - A^i_b\lambda^{aj}_{kl}$$
$$+A^i_k\lambda^{aj}_{bl} - A^i_k\lambda^a_l\xi^j_b - A^i_l\lambda^{aj}_{bk} + A^i_l\lambda^a_k\xi^j_b + A^j_b\lambda^{ai}_{kl} - A^j_k\lambda^{ai}_{bl} + A^j_k\lambda^a_l\xi^i_b$$
$$+A^j_l\lambda^{ai}_{bk} - A^j_l\lambda^a_k\xi^i_b + \lambda^{ai}_{kl}\xi^j_b + \lambda^{aij}_{bkl} - \lambda^{aj}_{kl}\xi^i_b - \lambda^a_k\lambda^{ij}_{bl} + \lambda^a_l\lambda^{ij}_{bk}$$

$$-A^{ai}_{kl}\lambda^j_b + A^{ai}_{kl}\xi^j_b + A^{aj}_{kl}\lambda^i_b - A^{aj}_{kl}\xi^i_b + A^{ij}_{bk}\lambda^a_l - A^{ij}_{bk}\xi^a_l - A^{ij}_{bl}\lambda^a_k + A^{ij}_{bl}\xi^a_k$$
$$-A^i_k\lambda^a_l\xi^j_b + A^i_k\lambda^j_b\xi^a_l + A^i_l\lambda^a_k\xi^j_b - A^i_l\lambda^j_b\xi^a_k + A^j_k\lambda^a_l\xi^i_b - A^j_k\lambda^i_b\xi^a_l - A^j_l\lambda^a_k\xi^i_b$$
$$+A^j_l\lambda^i_b\xi^a_k - \lambda^{ai}_{kl}\lambda^j_b + \lambda^{ai}_{kl}\xi^j_b + \lambda^{aj}_{kl}\lambda^i_b - \lambda^{aj}_{kl}\xi^i_b - \lambda^a_k\lambda^{ij}_{bl} + \lambda^a_l\lambda^{ij}_{bk} - \lambda^{ij}_{bk}\xi^a_l$$
$$+\lambda^{ij}_{bl}\xi^a_k$$

.cmt : [$\{A^a_b\},\{A^{ij}_{kl}\}$]= $\{A^a_b\}\{A^{ij}_{kl}\}$ -$\{A^{ij}_{kl}\}\{A^a_b\}$

applyRule()

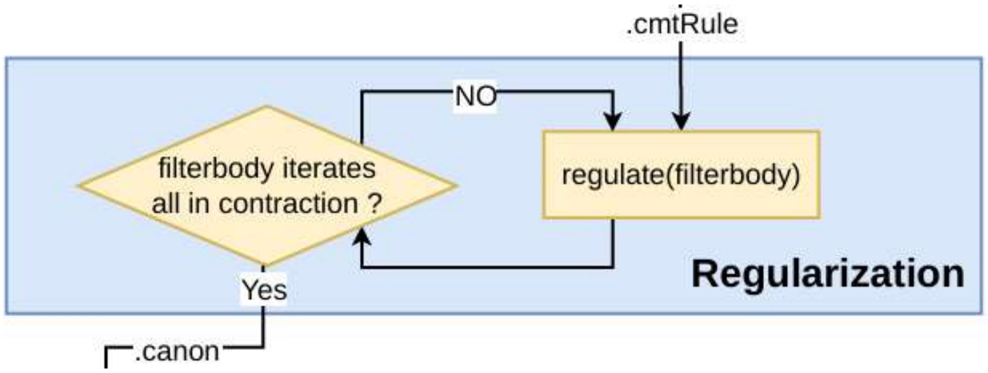
```
#diagonalize the xi and lambda
operators.applyRule(ruleType='both')
#the diagonalized commutator
jupyterDisplay(operators.cmtRule)

output:
```

$$-A^{ai}_{kl}\delta^j_b + A^{aj}_{kl}\delta^i_b + A^{ij}_{bk}\delta^a_l - A^{ij}_{bl}\delta^a_k + A^i_k n_a\delta^a_l\delta^j_b - A^i_k n_j\delta^a_l\delta^j_b - A^i_l n_a\delta^a_k\delta^j_b$$
$$+A^i_l n_j\delta^a_k\delta^j_b - A^j_k n_a\delta^a_l\delta^i_b + A^j_k n_i\delta^a_l\delta^i_b + A^j_l n_a\delta^a_k\delta^i_b - A^j_l n_i\delta^a_k\delta^i_b - \delta^a_k\lambda^{ij}_{bl}$$
$$+\delta^a_l\lambda^{ij}_{bk} + \delta^i_b\lambda^{aj}_{kl} - \delta^j_b\lambda^{ai}_{kl}$$

In the natural orbital basis(NAT), both $\lambda^{(1)}$ and $\xi^{(1)}$ matrices are diagonal.

$$\xi^i_j = \lambda^i_j - \delta^i_j.$$

$$\lambda^i_j = n_i\delta^i_j, \qquad \xi^i_j = -\bar{n}_i\delta^i_j = -(1-n_i)\delta^i_j,$$




```
#filter [1B,2B]-1B and normalize the indices
operators.regulate(filterbody=1)
#filtered terms with Matrix elements
jupyterDisplay(operators.filterTerms)
#contract the delta and canonicalize indices
jupyterDisplay(operators.canon)
```

output:

$$A^i_k G^a_b H^{ij}_{kl} n_a \delta^a_l \delta^j_b - A^i_k G^a_b H^{ij}_{kl} n_j \delta^a_l \delta^j_b - A^i_l G^a_b H^{ij}_{kl} n_a \delta^a_k \delta^j_b + A^i_l G^a_b H^{ij}_{kl} n_j \delta^a_k \delta^j_b$$
$$-A^j_k G^a_b H^{ij}_{kl} n_a \delta^a_l \delta^i_b + A^j_k G^a_b H^{ij}_{kl} n_i \delta^a_l \delta^i_b + A^j_l G^a_b H^{ij}_{kl} n_a \delta^a_k \delta^i_b - A^j_l G^a_b H^{ij}_{kl} n_i \delta^a_k \delta^i_b$$

$$A^a_b G^c_d H^{ad}_{bc} n_c - A^a_b G^c_d H^{ad}_{bc} n_d - A^a_b G^c_d H^{ad}_{cb} n_c + A^a_b G^c_d H^{ad}_{cb} n_d - A^a_b G^c_d H^{da}_{bc} n_c$$
$$+A^a_b G^c_d H^{da}_{bc} n_d + A^a_b G^c_d H^{da}_{cb} n_c - A^a_b G^c_d H^{da}_{cb} n_d$$

Regularization module :

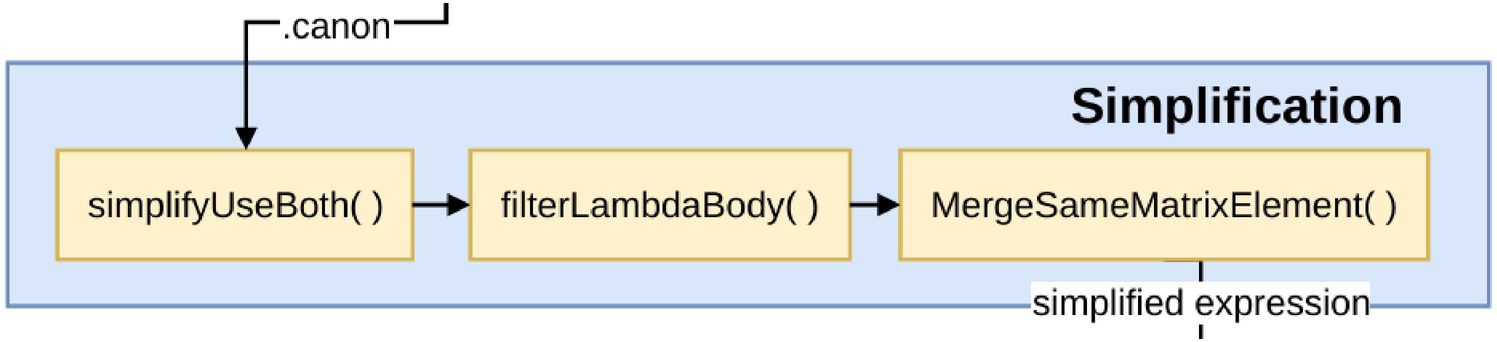
Compute $G^a_b H^{ij}_{kl}\left[\{A^a_b\},\left\{A^{ij}_{kl}\right\}\right]^{(k)}$

regulate()

- Multiplication by matrix elements of operators.
- Filtering by the properties of operators.
- Simplification and re-indexing.

.canon : $G^a_b H^{ij}_{kl}\left[\{A^a_b\}\left\{A^{ij}_{kl}\right\}\right]^{(1)}$

Simplification



Simplification module: Simplify the commutator expression.

simplifyUseBoth( )

- **simplifyUseAntisymmetry(expr)**

$$G^{ia}_{kc}H^{jb}_{ld} = (-1)^4 G^{ai}_{ck}H^{bj}_{dl}$$
$$-G^{ai}_{kc}H^{jb}_{ld} = -(-1)^3 G^{ai}_{ck}H^{bj}_{dl}.$$

- **simplifyUseDummyIndices(expr)**

$$A^a_b G^{cd}_{ef} H^{ag}_{bc} \lambda^{dg}_{ef} \rightarrow A^a_b G^{gc}_{ef} H^{ad}_{bg} \lambda^{cd}_{ef} \quad (\text{map } dgefc \rightarrow cdefg), \tag{33}$$

$$A^a_b G^{gd}_{ef} H^{ac}_{bg} \lambda^{dc}_{ef} \rightarrow A^a_b G^{gc}_{ef} H^{ad}_{bg} \lambda^{cd}_{ef} \quad (\text{map } dcefg \rightarrow cdefg). \tag{34}$$

```
#unsimplified expression
initial_expr = operators.canon
#simplify the expression
simplified_expr = qcombo.simplifyUseBoth(
    initial_expr)
jupyterDisplay(simplified_expr)

output:
```

$$4A^a_b G^c_d H^{ad}_{bc} n_c - 4A^a_b G^c_d H^{ad}_{bc} n_d$$

8 terms -> 2 terms!

Simplification

filterLambdaBody ( )

- Filter terms according to the body of the irreducible density matrices they contain

```
#filter 1B-lambda terms
filter_lambda_expr = qcombo.filterLambdaBody
    (simplified_expr,LambdaBody=1)
jupyterDisplay(filter_lambda_expr)

output:
```

$4A_b^aG_d^cH_{bc}^{ad}n_c - 4A_b^aG_d^cH_{bc}^{ad}n_d$

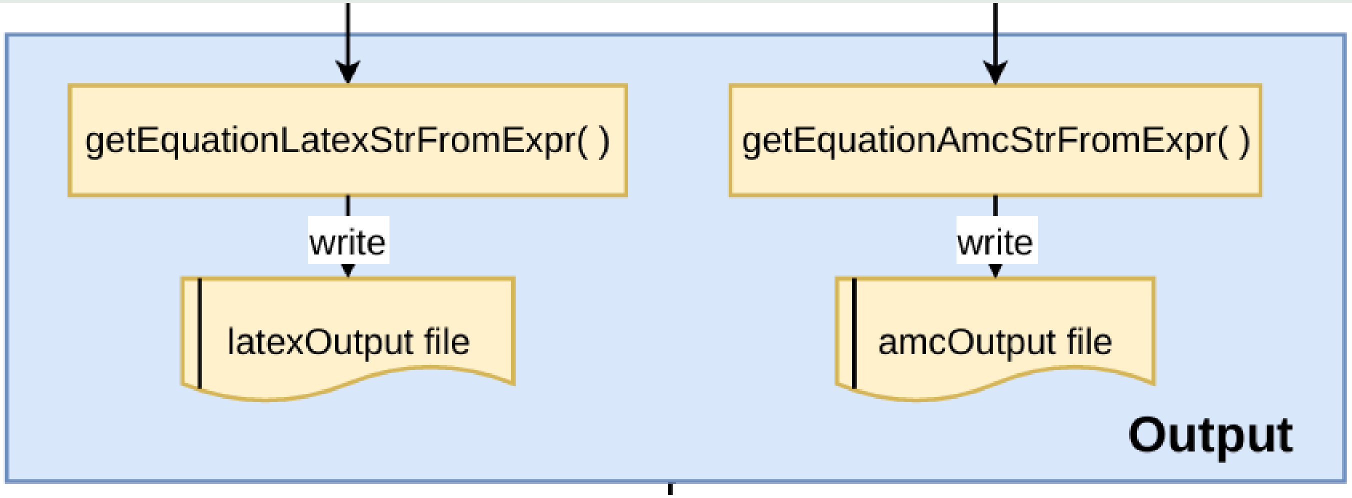
MergeSameMatrixElement( )

- Iterates over all terms and identifies those containing matrix elements with identical index structures.

```
#merge the same matrix element
united_expr = qcombo.MergeSameMatrixElement(
    filter_lambda_expr)
jupyterDisplay(united_expr)

output:
```

$4(n_c - n_d)A_b^aG_d^cH_{bc}^{ad}$

Output



Output module ： Write the simplified expression to Latex and amc syatnx.

getEquationLatexStrFromExpr( )

```
from qcombo.output import
    getEquationLatexStrFromExpr
#trans expresion to latex equation
latex_str = getEquationLatexStrFromExpr(
    united_expr,1,2,'R')
print(latex_str)

output:
R^{a}_{b} = \sum_{c d} (n^{}_{c}-n^{}_{d})
    \* G^{c}_{d} \* H^{ad}_{bc}
```

$$[G^{(1)}, H^{(2)}]^{(1)} = \sum_{ab} R^a_b\{A^a_b\}, \qquad R^a_b = \sum_{cd}(n_c - n_d)G^c_d H^{ad}_{bc}.$$

getEquationAmcStrFromExpr( )

```
from qcombo.output import
    getEquationAmcStrFromExpr
#trans expresion to amc equation
amc_str = getEquationAmcStrFromExpr(
    united_expr,1,2,'R')
print(amc_str)

output:
R1_ab = 1/4*sum_cd(4*(n_c-n_d)*G_cd*H_adbc);
```

Flowchart

Input

Commutator

$$\left[\{A^{a}_{b}\},\left\{A^{ij}_{kl}\right\}\right]$$

Regularization

$$G^{a}_{b}H^{ij}_{kl}\left[\{A^{a}_{b}\}\left\{A^{ij}_{kl}\right\}\right]^{(1)}$$

Simplification

Output

$$[G^{(1)}, H^{(2)}] = \frac{1}{4}\sum_{abijkl} G^{a}_{b}H^{ij}_{kl}\left[\{A^{a}_{b}\},\{A^{ij}_{kl}\}\right].$$

```
#[1B,2B]
left=[['a'],['b']]
right=[['i','j'],['k','l']]
```

$$-A^{ai}_{kl}\delta^{j}_{b} + A^{aj}_{kl}\delta^{i}_{b} + A^{ij}_{bk}\delta^{a}_{l} - A^{ij}_{bl}\delta^{a}_{k} + A^{i}_{k}n_{a}\delta^{a}_{l}\delta^{j}_{b} - A^{i}_{k}n_{j}\delta^{a}_{l}\delta^{j}_{b} - A^{i}_{l}n_{a}\delta^{a}_{k}\delta^{j}_{b}$$
$$+A^{i}_{l}n_{j}\delta^{a}_{k}\delta^{j}_{b} - A^{j}_{k}n_{a}\delta^{a}_{l}\delta^{i}_{b} + A^{j}_{k}n_{i}\delta^{a}_{l}\delta^{i}_{b} + A^{j}_{l}n_{a}\delta^{a}_{k}\delta^{i}_{b} - A^{j}_{l}n_{i}\delta^{a}_{k}\delta^{i}_{b} - \delta^{a}_{k}\lambda^{ij}_{bl}$$
$$+\delta^{a}_{l}\lambda^{ij}_{bk} + \delta^{i}_{b}\lambda^{aj}_{kl} - \delta^{j}_{b}\lambda^{ai}_{kl}$$

$$A^{a}_{b}G^{c}_{d}H^{ad}_{bc}n_{c} - A^{a}_{b}G^{c}_{d}H^{ad}_{bc}n_{d} - A^{a}_{b}G^{c}_{d}H^{ad}_{cb}n_{c} + A^{a}_{b}G^{c}_{d}H^{ad}_{cb}n_{d} - A^{a}_{b}G^{c}_{d}H^{da}_{bc}n_{c}$$
$$+A^{a}_{b}G^{c}_{d}H^{da}_{bc}n_{d} + A^{a}_{b}G^{c}_{d}H^{da}_{cb}n_{c} - A^{a}_{b}G^{c}_{d}H^{da}_{cb}n_{d}$$

$$4(n_{c} - n_{d})A^{a}_{b}G^{c}_{d}H^{ad}_{bc}$$

$$R^{a}_{b} = \sum_{cd}(n_{c} - n_{d})G^{c}_{d}H^{ad}_{bc}.$$

Latex

R1_ab = 1/4*sum_cd(4*(n_c-n_d)*G_cd*H_adbc);

AMC

easyCombo

To facilitate ease of use, we have integrated the steps described above into a unified function:

```
easyCombo(left, right, contraction=None,
    latexOutput=None, amcOutput=None, **
    kwargs).
```

1. `left`. The body of the left operator in the commutator. Accepts either an integer or a list of strings representing the operator indices, e.g., 1 or `[['a'],['b']]` both denote a normal-ordered one-body operator.
2. `right`. The body of the right operator in the commutator, with the same format as parameter `left`.
3. `contraction`. The desired contraction terms to output. Accepts an integer or a list of integers, e.g.,`0` or `[0, 1, ...]`. The default `None` outputs all possible contraction terms.
4. `latexOutput`. Filename for the LaTeX-formatted equation output. When set to `None` (default), the corresponding filename is automatically generated based on the `left`, `right`, and `contraction` parameters.
5. `amcOutput`. Filename for the amc-formatted equation output. When set to `None` (default), the corresponding filename is automatically generated based on the `left`, `right`, and `contraction` parameters.

The optional keyword arguments are as follows:

1. `wick_mode`. Specifies the reference-state mode for Wick contractions. The options are either `"SR"` or `"MR"` (default). The `"SR"` mode denotes a single-reference state, which restricts contractions to one-body irreducible density matrices. The `"MR"` mode denotes a multi-reference state, which permits contractions involving many-body irreducible density matrices.
2. `show_process`. The default is `True`. This flag determines whether the computation progress is displayed.
3. `parallel`. The default is `False`. This flag determines whether parallel computation is enabled.
4. `savefile`. The default is `True`. This flag determines whether the LaTeX output (as a `.tex` file) and the amc output (as a `.amc` file) are saved.

```
# [2B,2B]-0B
qcombo.easyCombo(2,2,0)
```

➢ Latex file

(1) one-body irreducible density matrix $\lambda^{(1)}$

$$R = \sum_{abcd}(-n_a n_b n_c - n_a n_b n_d + n_a n_b + n_a n_c n_d + n_b n_c n_d - n_c n_d) G^{ab}_{cd} H^{cd}_{ab}/4 \quad (37)$$

(2) two-body irreducible density matrix $\lambda^{(2)}$

$$R = \sum_{abcdef}\left[8(n_e - n_f) G^{ae}_{cf} H^{bf}_{de} - (n_e + n_f - 1) G^{ab}_{ef} H^{ef}_{cd} + (n_e + n_f - 1) G^{ef}_{cd} H^{ab}_{ef}\right] \lambda^{ab}_{cd}/8$$

(3) three-body irreducible density matrix $\lambda^{(3)}$

$$R = \sum_{abcdefg}(G^{ag}_{de} H^{bc}_{fg} - G^{ab}_{dg} H^{cg}_{ef}) \lambda^{abc}_{def}/4$$

➢ amc file

```
declare G{mode= (2,2),latex ="G" }
declare H{mode= (2,2),latex ="H" }
declare R0{mode= (0,0),latex ="R" }
declare lambda2B{mode= (2,2),latex ="\lambda
    " }
declare lambda3B{mode= (3,3),latex ="\lambda
    " }
declare n {  mode=2, diagonal=true, latex="n
    "}

# commutator [2B,2B]-0B
# lambda_1B
R0 = 1/16*sum_abcd(4*(-n_a*n_b*n_c-n_a*n_b*
    n_d+n_a*n_b+n_a*n_c*n_d+n_b*n_c*n_d-n_c*
    n_d)*G_abcd*H_cdab);

# lambda_2B
R0 = 1/16*sum_abcdef(2*(8*(n_e-n_f)*G_aecf*
    H_bfde-(n_e+n_f-1)*G_abef*H_efcd+(n_e+
    n_f-1)*G_efcd*H_abef)*lambda2B_abcd);

# lambda_3B
R0 = 1/16*sum_abcdefg(4*(-G_abdg*H_cgef+
    G_agde*H_bcfg)*lambda3B_abcdef);
```

- **Coefficients simplified**

(1) one-body irreducible density matrix $\lambda^{(1)}$

$$R = \sum_{abcd}(-n_a n_b n_c - n_a n_b n_d + n_a n_b + n_a n_c n_d + n_b n_c n_d - n_c n_d) G^{ab}_{cd} H^{cd}_{ab}/4 \quad (37)$$

(2) two-body irreducible density matrix $\lambda^{(2)}$

$$R = \sum_{abcdef}\left[8(n_e - n_f) G^{ae}_{cf} H^{bf}_{de} - (n_e + n_f - 1) G^{ab}_{ef} H^{ef}_{cd} + (n_e + n_f - 1) G^{ef}_{cd} H^{ab}_{ef}\right] \lambda^{ab}_{cd}/8$$

(3) three-body irreducible density matrix $\lambda^{(3)}$

$$R = \sum_{abcdefg}(G^{ag}_{de} H^{bc}_{fg} - G^{ab}_{dg} H^{cg}_{ef}) \lambda^{abc}_{def}/4$$

1) $n_a n_b \bar{n}_c \bar{n}_d - \bar{n}_c \bar{n}_d n_a n_b = n_a n_b - n_a n_b n_c - n_a n_b n_d + n_a n_c n_d + n_b n_c n_d - n_c n_d$

2) $n_a n_b - \bar{n}_a \bar{n}_b = n_a + n_b - 1$

$$[G^{(2)}, H^{(2)}]^{(0)} = \left[\frac{1}{4}\sum_{abcd} G^{ab}_{cd}\{A^{ab}_{cd}\}, \frac{1}{4}\sum_{ijkl} H^{ij}_{kl}\{A^{ij}_{kl}\}\right]^{(0)}$$

$$= \sum_{abcd}\frac{1}{4}(n_a n_b \bar{n}_c \bar{n}_d - \bar{n}_a \bar{n}_b n_c n_d) G^{ab}_{cd} H^{cd}_{ab}$$

$$+ \sum_{abcdef}\left[(n_e - n_f) G^{ae}_{cf} H^{bf}_{de} + \frac{1}{8}(n_e n_f - \bar{n}_e \bar{n}_f)\left(G^{ef}_{cd} H^{ab}_{ef} - G^{ab}_{ef} H^{ef}_{cd}\right)\right]\lambda^{ab}_{cd}$$

$$+ \sum_{abcdefg}\frac{1}{4}(G^{ag}_{dc} H^{bc}_{fg} - G^{ab}_{dg} H^{cg}_{ef})\lambda^{abc}_{def}$$

Bash environment

For direct command-line execution, users can compute commutators and generate corresponding output files using the command:

```
qcombo LEFT RIGHT [OPTIONS]
```

1. LEFT. The body of the left operator (as an integer) or its specific index list (provided as a Python list string).
2. RIGHT. The body of the right operator (as an integer) or its specific index list (provided as a Python list string).

1. `--contraction CONTRACTION, -c CONTRACTION` Specify the desired contraction body numbers. This can be a single integer (e.g., 0), a range (e.g., 0-2), a list (e.g., 0,1,2), or 'all' to calculate all possible contractions. Default is 'all'.
2. `--latex-output LATEX_OUTPUT, -lo LATEX_OUTPUT` Specify the filename for the LaTeX output. Defaults to an auto-generated name based on the input operators.
3. `--amc-output AMC_OUTPUT, -ao AMC_OUTPUT` Specify the filename for the AMC program input file. Defaults to an auto-generated name based on the input operators.
4. `--wick-mode {SR, MR}, -w {SR, MR}` The reference state for Wick contraction. "SR" for single-reference state or "MR" for multi-reference state. Default is "MR".
5. `--parallel, -p` Enable parallel computation.
6. `--no-process, -ns` Disable progress output during computation.
7. `--version, -v` Display the program's version number and exit.
8. `--interactive -i` Launch interactive mode. Note that running the command without arguments also enters this mode.
9. `--help -h` Show the help message and exit.

MR-IMSRG(3) flow equation:

Except for dE, the flow equations for other parts are truncated to the two-body term λ

(I.) dE:

$$dE/ds = \sum_{ab}(n_a - n_b)\eta^a_b f^b_a$$
$$+ \sum_{abcd}\frac{1}{4}(n_a n_b \bar{n}_c \bar{n}_d - \bar{n}_a \bar{n}_b n_c n_d)\eta^{ab}_{cd}\Gamma^{cd}_{ab}$$
$$+ \frac{1}{36}\sum_{abcdef}(n_a n_b n_c \bar{n}_d \bar{n}_e \bar{n}_f - \bar{n}_a \bar{n}_b \bar{n}_c n_d n_e n_f)\eta^{abc}_{def} W^{def}_{abc}$$
$$+ \sum_{abcd}\frac{1}{4}(d\Gamma^{ab}_{cd}(12) + d\Gamma^{ab}_{cd}(22) + d\Gamma^{ab}_{cd}(23, \lambda^{(1)}) + d\Gamma^{ab}_{cd}(33, \lambda^{(1)}))\lambda^{ab}_{cd}$$
$$+ \frac{1}{2}\sum_{abcd}\frac{1}{4}(d\Gamma^{ab}_{cd}(23, \lambda^{(2)}) + d\Gamma^{ab}_{cd}(33, \lambda^{(2)}))\lambda^{ab}_{cd}$$
$$+ \sum_{abcdef}\frac{1}{36}\frac{dW^{abc}_{def}}{ds}\lambda^{abc}_{def}$$

(II.) df: $$df^a_b/ds = df^a_b(11) + df^a_b(12) + df^a_b(22) + df^a_b(13) + df^a_b(23) + df^a_b(33)$$

[1B,1B]-1B:

$$df^a_b(11) = \sum_c (\eta^a_c f^c_b - \eta^c_b f^a_c)$$

[2B,2B]-1B:

$$\begin{aligned} df^a_b(22) =& \frac{1}{2}\sum_{cde}(n_c\bar{n}_d\bar{n}_e + \bar{n}_c n_d n_e)(\eta^{ac}_{de}\Gamma^{de}_{bc} - \eta^{de}_{bc}\Gamma^{ac}_{de}) \\ &+ \sum_{cdefg}\Big[\frac{1}{4}(\eta^{ag}_{ef}\Gamma^{cd}_{bg} - \eta^{cd}_{bg}\Gamma^{ag}_{ef}) + (\eta^{ac}_{eg}\Gamma^{dg}_{bf} - \eta^{cg}_{be}\Gamma^{ad}_{fg}) \\ &+ \frac{1}{2}(\eta^{ag}_{be}\Gamma^{cd}_{fg} - \eta^{ac}_{bg}\Gamma^{dg}_{ef}) + \frac{1}{2}(\eta^{cd}_{eg}\Gamma^{ag}_{bf} - \eta^{cg}_{ef}\Gamma^{ad}_{bg})\Big]\lambda^{cd}_{ef} \end{aligned}$$

[1B,3B]-1B:

$$df^a_b(13) = \sum_{cdefg}\frac{1}{2}(\eta^g_e W^{acd}_{bfg} - \eta^c_g W^{adg}_{bef})\lambda^{cd}_{ef} - \sum_{cdefg}\frac{1}{2}(f^g_e \eta^{acd}_{bfg} - f^c_g \eta^{adg}_{bef})\lambda^{cd}_{ef}$$

df

[2B,3B]-1B:

$$
\begin{aligned}
df_b^a(23) = & \sum_{cdef} \frac{1}{4}(n_c n_d \bar{n}_e \bar{n}_f - \bar{n}_c \bar{n}_d n_e n_f)\eta^{cd}_{ef} W^{aef}_{bcd} \\
& + \sum_{cdefgh} \Big[(n_g - n_h)\eta^{cg}_{eh} W^{adh}_{bfg} - \frac{1}{2}(n_g - n_h)(\eta^{ag}_{eh} W^{cdh}_{bfg} + \eta^{cg}_{bh} W^{adh}_{efg}) \\
& + \frac{1}{4}(n_g n_h - \bar{n}_g \bar{n}_h)(\eta^{gh}_{be} W^{acd}_{fgh} - \eta^{ac}_{gh} W^{dgh}_{bef}) + \frac{1}{8}(n_g n_h - \bar{n}_g \bar{n}_h)(\eta^{gh}_{ef} W^{acd}_{bgh} - \eta^{cd}_{gh} W^{agh}_{bef}) \Big] \lambda^{cd}_{ef} \\
& - \sum_{cdef} \frac{1}{4}(n_c n_d \bar{n}_e \bar{n}_f - \bar{n}_c \bar{n}_d n_e n_f)\Gamma^{cd}_{ef} \eta^{aef}_{bcd} \\
& - \sum_{cdefgh} \Big[(n_g - n_h)\Gamma^{cg}_{eh} \eta^{adh}_{bfg} - \frac{1}{2}(n_g - n_h)(\Gamma^{ag}_{eh} \eta^{cdh}_{bfg} + \Gamma^{cg}_{bh} \eta^{adh}_{efg}) \\
& + \frac{1}{4}(n_g n_h - \bar{n}_g \bar{n}_h)(\Gamma^{gh}_{be} \eta^{acd}_{fgh} - \Gamma^{ac}_{gh} \eta^{dgh}_{bef}) + \frac{1}{8}(n_g n_h - \bar{n}_g \bar{n}_h)(\Gamma^{gh}_{ef} \eta^{acd}_{bgh} - \Gamma^{cd}_{gh} \eta^{agh}_{bef}) \Big] \lambda^{cd}_{ef}
\end{aligned}
$$

[3B,3B]-1B:

$$
\begin{aligned}
df_b^a(33) = & \sum_{cdefg} (n_c n_d n_e \bar{n}_f \bar{n}_g + \bar{n}_c \bar{n}_d \bar{n}_e n_f n_g)(\eta^{afg}_{cde} W^{cde}_{bfg} - \eta^{cde}_{bfg} W^{afg}_{cde}) \\
& + \sum_{cdef} \sum_{ghi} \Big[\frac{1}{2}(n_g \bar{n}_h \bar{n}_i + \bar{n}_g n_h n_i)(\eta^{acg}_{ehi} W^{dhi}_{bfg} - \eta^{chi}_{beg} W^{adg}_{fhi}) \\
& + \frac{1}{4}(n_g \bar{n}_h \bar{n}_i + \bar{n}_g n_h n_i)((\eta^{ahi}_{beg} W^{cdg}_{fhi} - \eta^{acg}_{bhi} W^{dhi}_{efg}) + (\eta^{cdg}_{ehi} W^{ahi}_{bfg} - \eta^{chi}_{efg} W^{adg}_{bhi})) \\
& + \frac{1}{8}(n_g \bar{n}_h \bar{n}_i + \bar{n}_g n_h n_i)(\eta^{ahi}_{efg} W^{cdg}_{bhi} - \eta^{cdg}_{bhi} W^{ahi}_{efg}) + \frac{1}{24}(n_g n_h n_i + \bar{n}_g \bar{n}_h \bar{n}_i)(\eta^{acd}_{ghi} W^{ghi}_{bef} - \eta^{ghi}_{bef} W^{acd}_{ghi}) \Big] \lambda^{cd}_{ef}
\end{aligned}
$$

(III.) $d\Gamma$:

$$d\Gamma^{ab}_{cd}/ds = d\Gamma^{ab}_{cd}(12) + d\Gamma^{ab}_{cd}(22) + d\Gamma^{ab}_{cd}(13) + d\Gamma^{ab}_{cd}(23) + d\Gamma^{ab}_{cd}(33)$$

[1B,2B]-2B:

$$d\Gamma^{ab}_{cd}(12) = \sum_{e} \hat{P}_{ab}(f^{a}_{e}\eta^{be}_{cd} - \eta^{a}_{e}\Gamma^{be}_{cd}) + \hat{P}_{cd}(\eta^{e}_{c}\Gamma^{ab}_{de} - f^{e}_{c}\eta^{ab}_{de})$$

[2B,2B]-2B:

$$d\Gamma^{ab}_{cd}(22) = \sum_{ef} \left[\hat{P}_{ab}\hat{P}_{cd}(n_e - n_f)\eta^{ae}_{cf}\Gamma^{bf}_{de} + \frac{1}{2}(n_e n_f - \bar{n}_e\bar{n}_f)(\eta^{ef}_{cd}\Gamma^{ab}_{ef} - \eta^{ab}_{ef}\Gamma^{ef}_{cd})\right]$$

[1B,3B]-2B:

$$d\Gamma^{ab}_{cd}(13) = \sum_{ef}(n_e - n_f)(\eta^{e}_{f}W^{abf}_{cde} - f^{e}_{f}\eta^{abf}_{cde})$$

[2B,3B]-2B:

$$d\Gamma^{ab}_{cd}(23) = d\Gamma^{ab}_{cd}(23,\lambda^{(1)}) + d\Gamma^{ab}_{cd}(23,\lambda^{(2)})$$

$$\begin{aligned}d\Gamma^{ab}_{cd}(23,\lambda^{(1)}) =& \frac{1}{2}\sum_{efg}(n_e\bar{n}_f\bar{n}_g + \bar{n}_e n_f n_g)(\hat{P}_{cd}\eta^{fg}_{ce}W^{abe}_{dfg} - \hat{P}_{ab}\eta^{ae}_{fg}W^{bfg}_{cde})\\ &-\frac{1}{2}\sum_{efg}(n_e\bar{n}_f\bar{n}_g + \bar{n}_e n_f n_g)(\hat{P}_{cd}\Gamma^{fg}_{ce}\eta^{abe}_{dfg} - \hat{P}_{ab}\Gamma^{ae}_{fg}\eta^{bfg}_{cde})\end{aligned}$$

$$\begin{aligned}d\Gamma^{ab}_{cd}(23,\lambda^{(2)}) =& \sum_{efghi}\Big[\frac{1}{2}(\eta^{ab}_{gi}W^{efi}_{cdh} - \eta^{ei}_{cd}W^{abf}_{ghi}) + \frac{1}{4}(\hat{P}_{cd}\eta^{ef}_{ci}W^{abi}_{dgh} - \hat{P}_{ab}\eta^{ai}_{gh}W^{bef}_{cdi}) + \frac{1}{2}(\eta^{ef}_{gi}W^{abi}_{cdh} - \eta^{ei}_{gh}W^{abf}_{cdi})\\ &+ (\hat{P}_{cd}\eta^{ei}_{cg}W^{abf}_{dhi} - \hat{P}_{ab}\eta^{ae}_{gi}W^{bfi}_{cdh}) + \frac{1}{2}\hat{P}_{ab}\hat{P}_{cd}(\eta^{ai}_{cg}W^{bef}_{dhi} - \eta^{ae}_{ci}W^{bfi}_{dgh})\Big]\lambda^{ef}_{gh}\\ &-\sum_{efghi}\Big[\frac{1}{2}(\Gamma^{ab}_{gi}\eta^{efi}_{cdh} - \Gamma^{ei}_{cd}\eta^{abf}_{ghi}) + \frac{1}{4}(\hat{P}_{cd}\Gamma^{ef}_{ci}\eta^{abi}_{dgh} - \hat{P}_{ab}\Gamma^{ai}_{gh}\eta^{bef}_{cdi}) + \frac{1}{2}(\Gamma^{ef}_{gi}\eta^{abi}_{cdh} - \Gamma^{ei}_{gh}\eta^{abf}_{cdi})\\ &+ (\hat{P}_{cd}\Gamma^{ei}_{cg}\eta^{abf}_{dhi} - \hat{P}_{ab}\Gamma^{ae}_{gi}\eta^{bfi}_{cdh}) + \frac{1}{2}\hat{P}_{ab}\hat{P}_{cd}(\Gamma^{ai}_{cg}\eta^{bef}_{dhi} - \Gamma^{ae}_{ci}\eta^{bfi}_{dgh})\Big]\lambda^{ef}_{gh}\end{aligned}$$

[3B,2B]-2B: $d\Gamma^{ab}_{cd}(33) = d\Gamma^{ab}_{cd}(33,\lambda^{(1)}) + d\Gamma^{ab}_{cd}(33,\lambda^{(2)})$

$$\begin{aligned} d\Gamma^{ab}_{cd}(33,\lambda^{(1)}) = \sum_{efgh} \Big[&\frac{1}{4}\hat{P}_{ab}\hat{P}_{cd}(n_e n_f \bar{n}_g \bar{n}_h - \bar{n}_e \bar{n}_f n_g n_h)\eta^{aef}_{cgh} W^{bgh}_{def} \\ &+ \frac{1}{6}(n_e \bar{n}_f \bar{n}_g \bar{n}_h - \bar{n}_e n_f n_g n_h)(\eta^{abe}_{fgh} W^{fgh}_{cde} - \eta^{fgh}_{cde} W^{abe}_{fgh}) \Big] \end{aligned}$$

$$\begin{aligned} d\Gamma^{ab}_{cd}(33,\lambda^{(2)}) &= \sum_{efgh}\sum_{ij} \Big[(n_i - n_j)\hat{P}_{ab}\hat{P}_{cd}\eta^{aei}_{cgj} W^{bfj}_{dhi} - \frac{1}{2}(n_i - n_j)(\hat{P}_{cd}(\eta^{abi}_{cgj} W^{efj}_{dhi} + \eta^{efi}_{cgj} W^{abj}_{dhi}) \\ &+ \hat{P}_{ab}(\eta^{aei}_{cdj} W^{bfj}_{ghi} + \eta^{aei}_{ghj} W^{bfj}_{cdi})) + \frac{1}{4}(n_i - n_j)(\eta^{abi}_{ghj} W^{efj}_{cdi} + \eta^{efi}_{cdj} W^{abj}_{ghi}) \\ &+ \frac{1}{2}(n_i n_j - \bar{n}_i \bar{n}_j)((\eta^{eij}_{cdg} W^{abf}_{hij} - \eta^{abe}_{gij} W^{fij}_{cdh}) + \frac{1}{2}(\hat{P}_{cd}\eta^{eij}_{cgh} W^{abf}_{dij} - \hat{P}_{ab}\eta^{aef}_{gij} W^{bij}_{cdh}) \\ &+ \frac{1}{2}(\hat{P}_{ab}\eta^{aij}_{cdg} W^{bef}_{hij} - \hat{P}_{cd}\eta^{abe}_{cij} W^{fij}_{dgh}) + \frac{1}{4}\hat{P}_{ab}\hat{P}_{cd}(\eta^{aij}_{cgh} W^{bef}_{dij} - \eta^{aef}_{cij} W^{bij}_{dgh})) \Big] \lambda^{ef}_{gh} \end{aligned}$$

(IV.) dW: $$dW^{abc}_{def}/ds = dW^{abc}_{def}(22) + dW^{abc}_{def}(13) + dW^{abc}_{def}(23) + dW^{abc}_{def}(33)$$

[2B,2B]-3B: $$dW^{abc}_{def}(22) = \sum_g (\hat{P}(a/bc)\hat{P}(de/f)(\eta^{ag}_{de}\Gamma^{bc}_{fg} - \eta^{bc}_{fg}\Gamma^{ag}_{de})$$

[1B,3B]-3B: $$dW^{abc}_{def}(13) = \sum_g \hat{P}(a/bc)(\eta^a_g W^{bcg}_{def} - f^a_g \eta^{bcg}_{def}) + \hat{P}(d/ef)(f^g_d \eta^{abc}_{efg} - \eta^g_d W^{abc}_{efg})$$

[2B,3B]-3B: $$dW^{abc}_{def}(23) = \sum_{gh} \Big[(n_g - n_h)\hat{P}(a/bc)\hat{P}(d/ef)(\eta^{ag}_{dh} W^{bch}_{efg} - \Gamma^{ag}_{dh}\eta^{bch}_{efg}) + \frac{1}{2}(n_g n_h - \bar{n}_g \bar{n}_h)(\hat{P}(de/f)(\eta^{gh}_{de} W^{abc}_{fgh} - \Gamma^{gh}_{de}\eta^{abc}_{fgh}) - \hat{P}(ab/c)(\eta^{ab}_{gh} W^{cgh}_{def} - \Gamma^{ab}_{gh}\eta^{cgh}_{def})) \Big]$$

dW

[3B,3B]-3B: $dW^{abc}_{def}(33) = dW^{abc}_{def}(33,\lambda^{(1)}) + dW^{abc}_{def}(33,\lambda^{(2)})$

$$dW^{abc}_{def}(33,\lambda^{(1)}) = \sum_{ghi}\Big[\frac{1}{2}(n_g\bar{n}_h\bar{n}_i + \bar{n}_g n_h n_i)(\hat{P}(ab/c)\hat{P}(d/ef)\eta^{abg}_{dhi}W^{chi}_{efg} - \hat{P}(a/bc)\hat{P}(de/f)\eta^{ahi}_{deg}W^{bcg}_{fhi})$$
$$+\frac{1}{6}(n_g n_h n_i + \bar{n}_g\bar{n}_h\bar{n}_i)(\eta^{abc}_{ghi}W^{ghi}_{def} - \eta^{ghi}_{def}W^{abc}_{ghi})\Big]$$

$$dW^{abc}_{def}(33,\lambda^{(2)}) = \sum_{ghij}\sum_{k}\Big[(\hat{P}(a/bc)\hat{P}(de/f)\eta^{agk}_{dei}W^{bch}_{fjk} - \hat{P}(ab/c)\hat{P}(d/ef)\eta^{abg}_{dik}W^{chk}_{efj})$$
$$+\frac{1}{2}(\hat{P}(ab/c)\hat{P}(de/f)(\eta^{abk}_{dei}W^{cgh}_{fjk} - \eta^{abg}_{dek}W^{chk}_{fij}) + \hat{P}(a/bc)\hat{P}(d/ef)(\eta^{agh}_{dik}W^{bck}_{efj} - \eta^{agk}_{dij}W^{bch}_{efk}))$$
$$+\frac{1}{4}(\hat{P}(a/bc)\hat{P}(de/f)\eta^{agh}_{dek}W^{bck}_{fij} - \hat{P}(ab/c)\hat{P}(d/ef)\eta^{abk}_{dij}W^{cgh}_{efk})$$
$$+\frac{1}{2}(\hat{P}(d/ef)\eta^{abc}_{dik}W^{ghk}_{efj} + \hat{P}(de/f)\eta^{ghk}_{dei}W^{abc}_{fjk} - \hat{P}(ab/c)\eta^{abg}_{ijk}W^{chk}_{def} - \hat{P}(a/bc)\eta^{agk}_{def}W^{bch}_{ijk})$$
$$+\frac{1}{4}(\eta^{abc}_{ijk}W^{ghk}_{def} - \eta^{ghk}_{def}W^{abc}_{ijk})\Big]\lambda^{gh}_{ij}$$

Summary

- 多体算符的对易式计算对于量子多体方法来说非常重要，而随着多体项的增加，人为计算对易式变得繁琐困难。因此我们基于Python的Sympy库，发展出了计算多体算符对易式的qcombo软件包，实现自动计算并化简正规序多体算符的对易式。借助qcombo程序，我们推导得到了MR-IMSRG(3)的流方程表达式。

- 我们将qcombo与角动量耦合程序amc对接，便于得到J-scheme下的对易式表达式。

Outlook

- 目前qcombo只支持计算粒子数守恒算符的对易式表达式，下一步将其拓展至粒子数破缺算符的对易式计算。

Thanks for your attention!